

Foundational Research & Advances in Iterative Quantum Phase Estimation (IQPE)

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Iterative Quantum Phase Estimation (IQPE) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Standard phase estimation needs a dozen quantum qubits working at the same time. Iterative phase estimation uses just ONE helper qubit, measuring one binary digit at a time and tweaking the knob for the next digit."

3. Mathematical Formulation & Unitary Rule

$$R_z(-\omega_k) \left[\frac{|0\rangle + e^{2\pi i (2^{k-1}\theta)}|1\rangle}{\sqrt{2}} \right]$$

Single-ancilla phase feedback cancels lower-order binary fraction phases, mapping the k-th bit directly to {0, 1}.

4. Step-by-Step Circuit Sequence

Single ancilla qubit, controlled- $U^{2^{k-1}}$, classical-controlled $R_z(-\omega_k)$ feedback rotation, Hadamard, measurement, reset.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit
from qiskit.circuit.library import PhaseGate

# Target qubit in |1>, 1 ancilla qubit
qc = QuantumCircuit(2, 2)
qc.x(1) # Target is eigenstate of T gate with theta = 0.001 (binary)

# Step 1: Extract lowest bit
qc.h(0)
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
$O(2^m)$ classical matrix diagonalization	1 ancilla qubit + m sequential iterations	Exponential with $O(m)$ total gates

7. Key Applications & Future Research Directions

- Molecular ground state energy estimation on NISQ quantum hardware
- Precision quantum metrology with constrained physical qubit counts
- Low-overhead eigenvalue extraction for fault-tolerant arithmetic

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Algorithms pipelines.